

1. Why do researchers create AI systems that play games?

1. The hardness of games

- i. Russel et al. said one of the characteristics of games, that is, hardness. They found that some games are too hard to solve, in other words, too time-complex to solve within reasonable time. For example, chess is extremely hard for an AI player to find an optimal solution. To search for an optimal solution in one chess game by brute force way, almost 5^{100} nodes have to be searched. It takes too much time for an AI player to visit all of them. So, AI researchers have developed a number of interesting ideas to make close-to-optimal solution within reasonable time [1].

2. The easiness for modeling

- i. Russel et al. evaluated the easiness of the modeling of the computer games. They said that the simplicity of computer games modeling makes them appealing to AI researchers. For example, while physical games like croquet have much more complex game states and actions to consider, the stage of computer game is easier to express, and the actions are restricted to a small number [1].

2. What are three possible application areas for AI systems that play games?

1. Finance trading robots

- i. Bikash explained how the AI game playing techniques can be utilized in the stock market area. He said that AI bots can compete with one another player (human or another AI bots) in milliseconds on the stock market. He also said that the AI models can help simulate adversarial conditions and adjust strategies based on competitors because the model is based on game theory [2].

2. Cybersecurity

- i. Bikash said that one of the application areas of game playing AI is cybersecurity. The game playing AI systems can detect anomalies and predict intrusion strategies based on game theory [2].

3. Autonomous systems

- i. Bikash said that another application areas of game playing AI is autonomous vehicles. He said that the navigation of vehicles needs strategic decisions. So, game playing AI's algorithms can be used to react to nearby vehicle behaviors by search for the simulation-based decision

trees [2].

3. What do researchers think are the ethical problems with game playing AIs?

1. Rodrigo et al. pointed out the fairness issue of the game playing AI versus human competition. They said that there is no fairness comparison between AI and a human. They explained the unfairness concept as this. "In the realm of sports and gaming, a common stance seems to be that a competition is unfair if one side is substantially favored by circumstances that should not affect the game" [3]. they said 7 fairness dimensions to determine the fairness of the competition as below.

- i. Input fairness

1. Do the two competitors have the same input space? For example, is the image capturing apparatus of AI equivalent to a human eye?

- ii. Output fairness

1. Are the two competitors limited by the same constraints of strength, speed and precision? For example, the AI has the same reaction speed?

- iii. Experience fairness

1. have the two competitors played the game under the same conditions?

- iv. Knowledge fairness

1. Do the two competitors have the same access to the same knowledge about the game?

- v. Compute fairness

1. Do the two competitors have the same computational power? The metric would be the power consumption in watts or total energy in joules. But in the case of humans, it is not easy to determine power is used

- vi. Psychological fairness

1. The factor that only apply to only humans

- vii. Common sense fairness

1. Do the two competitors have the same common-sense knowledge?

2. As a result, Rodrigo et al. stated that according to above the 6 fairness dimensions, the competition between current AI and human is unfair. They said that the complete fairness in all dimensions would require the system that is essentially equivalent to a read flash and blood human with a full history of huma-like experiences [3].

4. Are neural networks the best game players?

1. The answer is Yes

2. The evidence for my "Yes" answer

- i. Adria et al. proposed Agent57, which is the first deep Reinforcement Learning agent that outperforms the standard human benchmark on all 57 Atari games. They said that they trained a recurrent neural network for their new NGU agent [4]. So neural network is used in one of the state-of-the-art game playing AI.
- ii. Kai et al. proposed Alphastar, which beat a professional human player at the StarCraft 2 game. They said that Alphastar is a neural network-based AI system [5]. So neural network is also used in another kind of strong game playing AI

5. How reliable are my references?

1. Artificial Intelligence: A Modern Approach [1]

- i. This is the textbook of Artificial Intelligence. The reference count is 1054. very reliable reference

1. The evidence of the reference count

- a. [https://scholar.google.co.kr/scholar?q=Russell,+stuart+artificial+intelligence+a+modern+approach+\(2002\)&hl=ko&as_sdt=0&as_vis=1&oi=scholar](https://scholar.google.co.kr/scholar?q=Russell,+stuart+artificial+intelligence+a+modern+approach+(2002)&hl=ko&as_sdt=0&as_vis=1&oi=scholar)

2. Agent57: Outperforming the Atari Human Benchmark [4]

- i. The reference count is 771. So this paper is highly reliable.

1. The evidence of the reference count

- a. https://scholar.google.co.kr/scholar?hl=ko&as_sdt=0%2C5&as_vis=1&q=Agent57%3A+Outperforming+the+Atari+Human+Benchmark&btnG=

3. AlphaStar: An Evolutionary Computation Perspective [5]

- i. The reference count is 320. So, it is highly reliable reference paper.

1. The evidence of the reference count

- a. https://scholar.google.co.kr/scholar?hl=ko&as_sdt=0%2C5&as_vis=1&q=AlphaStar%3A+An+Evolutionary+Computation+Perspective&btnG=

6. References

- [1] Stuart Russel, Peter Norvig. 2010. Artificial Intelligence: A Modern Approach Third Edition, Prentice Hall
- [2] Bikash Daga. 2025, Game Theory Meets AI: Real-World Applications of AI Game Playing in 2025. <https://dev.to/bikashdaga/game-theory-meets-ai-real-world-applications-of-ai-game-playing-in-2025-2am2>
- [3] Rodrigo Canaan, Christoph Salge, Julian Togelius, Andy Nealen. 2019. Leveling the Playing Field – Fairness in AI Versus Human Game Benchmarks. ArXiv:1930.07008 (2020)

- [4] Adria Puigdomenech Badia, Bilal Piot, Steven Kapturowski et al. 2020. Agent57: Outperforming the Atari Human Benchmark, arXiv: 2003.13350(2020)
- [5] Kai Arulkumaran, Antoine Cully, Julian Togelius. 2019. AlphaStar: An Evolutionary Computation Perspective. arXiv:1902.01724 (2019)